

★ MDE >> observed difference across all pairs → 'no detectable difference' is statistically well-powered

★ MDE (Minimum Detectable Effect)

The threshold 'X' where:
 "If a true difference $\geq X$ exists,
 we detect it with 95% probability"

$$\text{MDE} = 1.96 \times \sqrt{(\text{SE}_1^2 + \text{SE}_2^2)}$$

$\uparrow$ $\alpha=0.05$ $\uparrow$ Bootstrap SE

Decision Criteria:
 obs < MDE → 'NS'
 = No diff, despite power to detect
 = Evidence of true equivalence

obs > MDE → 'Significant'
 = Significant diff detected

Why MDE matters:
 Standard $p > 0.05$ only means:
 "Failed to find a difference"
 $\neq$ "No difference exists"

MDE proves 'not underpowered',
 validating the claim of
 universality.

